DSBA 6010 – Gradient Descanters – Milestone 13: Deploy & Test

Overview & Functionality:

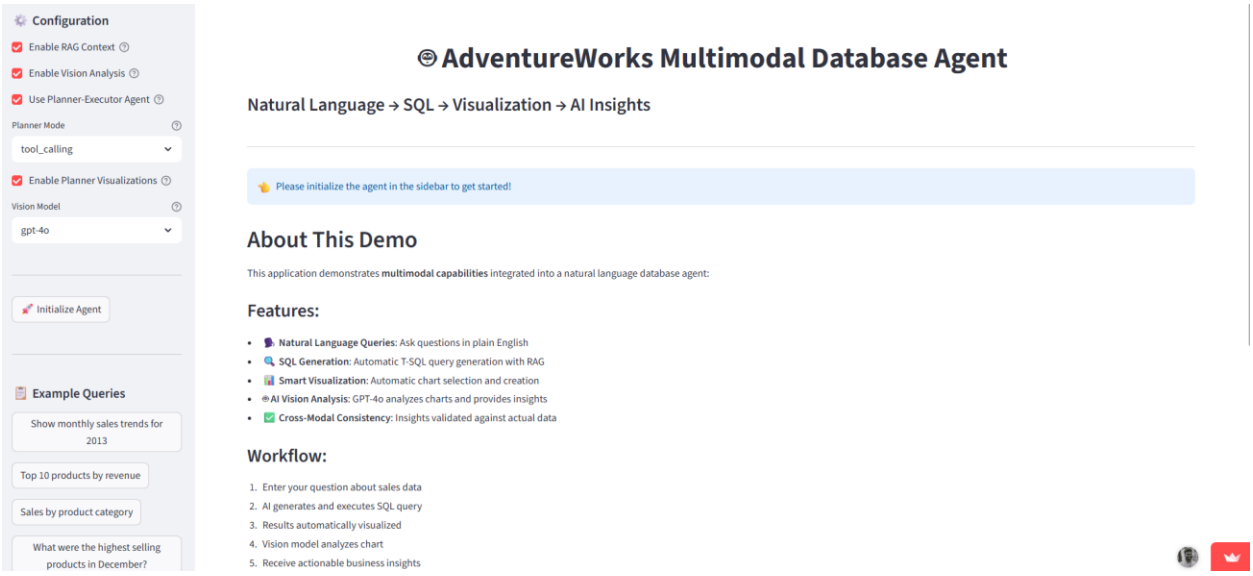
Our Database Agent application has been deployed on the streamlit cloud at the following link, https://gradientdescentersdatabaseagent.streamlit.app/ and is fully functional. Users can use the multimodal agent to query the AdventureWorks database in natural language, and receive back accurate SQL queries, along with the table results of the query, a dynamic visualization appropriate to the SQL query, and a vision analysis that breaks down key insights from the table and data visualization. All of this is drawn from a RAG system which loads documents into a vector database, with join patterns, table schemas, business rules, and privacy rules. We also have enabled tool calling, as well as a planner executor agent so that the application can plan and execute the user's natural language queries with the right guidelines.

We have also implemented monitoring and logging systems that you can see directly in the UI, shown under the "Agent Trace" and "Performance Metrics" tab which show all the steps which the agent is taking, as well as metrics on latency and token usage.

We have also included a query history section, which shows all the previous natural language queries which the user has requested, so that they can look back at previous history and results. Finally, in the case of errors, we have included the functionality for the error messages to be displayed on the screen to help with debugging.

The relevant screenshots displaying all this functionality is shown in the appendix below.

Appendix:



Configuration

- [x] Enable RAG Context
- [x] Enable Vision Analysis
- [x] Use Planner-Executor Agent

Planner Mode

tool_calling

- [x] Enable Planner Visualizations

Vision Model

gpt-4o

Initialize Agent

Agent initialized successfully!

AdventureWorks Multimodal Database Agent

Natural Language → SQL → Visualization → AI Insights

Ask a question about the sales data:

Compare sales across different regions in 2013

Process Query

Advanced Options

- [x] Generate Visualization
- [x] Enable Vision Analysis

Force Chart Type (optional)

Auto

- [x] Show Generated SQL
- [x] Show Agent Trace

Generated SQL

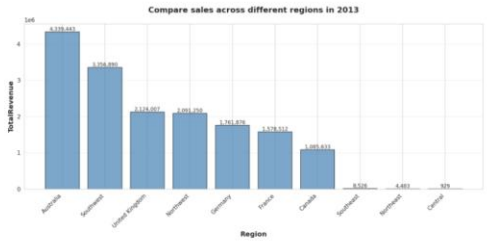
```
SELECT
    st.SalesTerritoryRegion AS Region,
    SUM(fis.SalesAmount) AS TotalRevenue,
    COUNT(DISTINCT fis.SalesOrderNumber) AS OrderCount
FROM dbo.FactInternetSales fis
INNER JOIN dbo.DimSalesTerritory st ON fis.SalesTerritoryKey = st.SalesTerritoryKey
INNER JOIN dbo.DimDate d ON fis.OrderDateKey = d.DateKey
WHERE d.CalendarYear = 2013
GROUP BY st.SalesTerritoryRegion
ORDER BY TotalRevenue DESC;
```

Query Results

Returned 10 rows in 13.69s

	Region	TotalRevenue	OrderCount
0	Australia	4339443.38	4640
1	Southwest	3356890.1	4319
2	United Kingdom	2124007.29	2377
3	Northwest	2091249.51	3242
4	Germany	1761876.36	1909
5	France	1578511.8	1917
6	Canada	1085632.65	2856
7	Southeast	8526.47	12
8	Northeast	4483.37	9
9	Central	929.41	8

Chart Type: bar



AI Visual Insights

Leaders: - **Australia**: $4,339,443 - **Southwest**: $3,356,890
Laggards:

- **Central**: $929
- **Northeast**: $4,483
- **Southeast**: $8,526

Distribution:

- The revenue is heavily concentrated in a few regions, with Australia and Southwest contributing significantly more than others.
- The majority of regions have revenues below the average of $1,635,155, indicating a skewed distribution.

Insights:

1. **Top Performers' Contribution**: Australia and Southwest together contribute approximately 47% of the total revenue, highlighting their importance to the business.
2. **Underperforming Regions**: Central, Northeast, and Southeast have negligible contributions. These regions may need targeted marketing strategies or product offerings to boost sales.
3. **Mid-Performers**: United Kingdom, Northwest, Germany, and France have moderate performance. Exploring successful strategies from top regions could help improve their sales.
4. **Market Expansion**: Consider investigating the high-performing regions to understand customer preferences and replicate successful strategies in underperforming areas.
5. **Resource Allocation**: Allocate more resources and marketing efforts to potential growth regions like United Kingdom and Northwest, while reassessing the viability of operations in Central, Northeast, and Southeast.

By focusing on these insights, AdventureWorks can optimize its sales strategies and improve overall performance across regions.

Vision tokens used: 2051

Agent Trace

step	name	status	error
1	Retrieve context	completed	None
2	Generate SQL	completed	None
3	Validate SQL	completed	None
4	Execute SQL	completed	None
5	Generate visualization	completed	None
6	Analyze visualization	completed	None

Performance Metrics

SQL Generation	Total Time	Total Tokens
2.93s	15.94s	3822

Query History

1. Show me a breakdown of sales across subcategories within the Bikes category, specific to Australia in the year 2013

2. Show me our top 5 customers in the United Kingdom. Who are they?

Status: ✅ Success

Rows: 5

Time: 15.82s

AI Insights: ✅ Generated

3. Compare sales across different regions in 2013

❌ An error occurred: 'NoneType' object has no attribute 'get'